FERNA SUNDAREE

Installation Guide for Linux

Install, update and remove Ferna Sundaree on Ubuntu, Debian, Fedora, openSUSE and other distributions, on x64 and ARM64.

Version 0.6.2 · November 2026



Ferna Software Solutions

Ferna Sundaree is free, open-source software (MIT licence)

github.com/tgi25/ferna-sundaree

© 2026 TGI Fernando, Ferna Software Solutions. Windows is a trademark of Microsoft Corporation; Linux is a trademark of Linus Torvalds. Other names belong to their owners.

Contents

1. Before you start 2
2. Install the .deb package (Ubuntu, Debian, Mint) 3
3. Install the .rpm package (Fedora, openSUSE) 4
4. Use the AppImage 4
5. Use the .tar.gz archive 6
6. After installing 6
7. System components some features use 8
8. Updates 8
9. Uninstall 9
10. For schools and organisations 10
11. Troubleshooting 11

Quick start for Ubuntu, Debian and Linux Mint

1. Download **FernaSundaree-0.6.2-x64.deb** (or **-arm64.deb** on ARM) from github.com/tgi25/ferna-sundaree/releases.
2. In a terminal in your Downloads folder, run `sudo apt install ./FernaSundaree-0.6.2-x64.deb`
3. Start **Ferna Sundaree** from your applications menu.
4. Optional: *Menu* → *Settings* → *General* → *Make default*.

Fedora: `sudo dnf install ./FernaSundaree-0.6.2-x64.rpm`. Other distributions: see Chapter 4 (AppImage) or Chapter 5 (.tar.gz).

CHAPTER 1

Before you start

This guide installs Ferna Sundaree 0.6.2 on Linux, keeps it up to date and removes it again. It also covers the optional system components that some features use, and settings for schools and offices.

System requirements

Item	Requirement
Processor	64-bit Intel/AMD (**x64**) or 64-bit ARM (**ARM64**, for example a Raspberry Pi 4/5 with a 64-bit OS, or a Snapdragon laptop)
Distribution	A current desktop distribution. Packages are provided for Debian and Ubuntu and their relatives (Linux Mint, Pop!_OS, elementary OS, Zorin OS, 64-bit Raspberry Pi OS), for Fedora, RHEL and openSUSE, and for any other distribution (AppImage, .tar.gz). This version was tested on Ubuntu 24.04 LTS.
Desktop	GNOME, KDE Plasma, Xfce, Cinnamon, MATE or similar, on Wayland or X11
Memory	4 GB RAM or more (8 GB for many tabs at once)
Disk space	About 350 MB for the program, plus space for your browsing data
Rights	Installing the .deb or .rpm needs `sudo`. The AppImage and .tar.gz run without it.

Find your processor type

Open a terminal (in most desktops Ctrl + Alt + T) and run:

```
uname -m
```

`x86_64` means you need the **x64** files. `aarch64` means you need the **arm64** files.

Choose a package

File (x64 shown)	Use it on	Updates
FernaSundaree-0.6.2-x64.deb Recommended	Ubuntu, Debian, Linux Mint, Pop!_OS, elementary OS, Zorin OS, Raspberry Pi OS. Adds a menu entry, registers the browser and keeps Chromium's security sandbox working on Ubuntu 23.10 and later.	Automatic; asks for your password once per update
FernaSundaree-0.6.2-x64.rpm	Fedora, RHEL and its rebuilds (AlmaLinux, Rocky), openSUSE	Automatic; asks for your password

FernaSundaree-0.6.2-x64.AppImage	Any distribution, without installing. Not recommended on Ubuntu 23.10 or later (see Chapter 4).	Automatic; the file replaces itself
FernaSundaree-0.6.2-x64.tar.gz	Any distribution, unpacked into a folder of your choice	The new archive is saved for you to unpack

For ARM computers, choose the same file with **arm64** instead of x64, for example `FernaSundaree-0.6.2-arm64.deb`.

Download and check the files

1. Open **github.com/tgi25/ferna-sundaree/releases**. Under the newest version, open **Assets** and download your package and **SHA256SUMS.txt**.
2. In a terminal, go to the folder and check the checksum:

   ```
   cd ~/Downloads
   sha256sum -c --ignore-missing SHA256SUMS.txt
   ```

   The line for your file must end in **OK**. If it says **FAILED**, delete the file and download it again. Never install a file that fails the check.

CHAPTER 2

Install the .deb package (Ubuntu, Debian, Mint)

This is the recommended way on Debian-based systems.

1. Open a terminal in the folder with the file and run (use the arm64 file on ARM):

   ```
   sudo apt install ./FernaSundaree-0.6.2-x64.deb
   ```

   The `./` matters: it tells apt to install this file. apt also installs any libraries that are missing.
2. Type your password when asked, and answer **Y** to continue.
3. Start **Ferna Sundaree** from your applications menu, or run `ferna-sundaree` in a terminal.

On many desktops you can also double-click the .deb file to open it in a software installer. The terminal command works everywhere and shows clear messages if something is missing.

What the .deb installs

- The browser in `/opt/Ferna Sundaree` and the command `/usr/bin/ferna-sundaree`.
- A menu entry with the Ferna Sundaree icon, registered as a web browser and as a viewer for PDF, HTML and `.fernaexam` files.

- On Ubuntu 23.10 and later, an AppArmor profile that lets Chromium's sandbox work. On older systems, the sandbox helper is set up instead.
- Example policy files for administrators in `/opt/Ferna Sundaree/resources/policy`.

CHAPTER 3

Install the .rpm package (Fedora, openSUSE)

Fedora, RHEL, AlmaLinux, Rocky Linux

```
sudo dnf install ./FernaSundaree-0.6.2-x64.rpm
```

openSUSE Leap and Tumbleweed

```
sudo zypper install --allow-unsigned-rpm ./FernaSundaree-0.6.2-x64.rpm
```

The packages are protected by the checksum and by Ferna's signed update list, not by a distribution GPG key. That is why zypper needs `--allow-unsigned-rpm`. If your dnf version refuses the file because it is not signed, check the checksum (Chapter 1) and add `--nogpgcheck`.

The .rpm installs the same files as the .deb: `/opt/Ferna Sundaree`, the `ferna-sundaree` command and the menu entry.

CHAPTER 4

Use the AppImage

An AppImage is a single file that runs without installation. Keep it somewhere permanent, for example in `~/Applications`.

1. Make it executable and run it:

   ```
   mkdir -p ~/Applications
   mv ~/Downloads/FernaSundaree-0.6.2-x64.AppImage ~/Applications/
   chmod +x ~/Applications/FernaSundaree-0.6.2-x64.AppImage
   ~/Applications/FernaSundaree-0.6.2-x64.AppImage
   ```

   You can also right-click the file → *Properties* → *Permissions* → *Allow executing file as program*, then double-click it.

2. If nothing happens or you see an error about **FUSE**, install FUSE 2:

```
# Ubuntu 24.04 and later
sudo apt install libfuse2t64
# Ubuntu 22.04, Debian 12, Linux Mint 21
sudo apt install libfuse2
# Fedora
sudo dnf install fuse-libs
```

3. To add it to your applications menu and make it your default browser, open *Menu* → *Settings* → *General* and click **Make default**. This creates a menu entry for the AppImage in `~/.local/share/applications`.

> **Ubuntu 23.10, 24.04 and later: use the .deb instead**
>
> Newer Ubuntu releases restrict the kernel feature that Chromium's security sandbox needs, unless an AppArmor profile allows it. The .deb installs such a profile; an AppImage cannot. Ferna Sundaree never turns its sandbox off, so on these Ubuntu releases the AppImage stops with a message about the sandbox. **Use the .deb package**, or the .tar.gz with the step in Chapter 5. Never start the browser with `--no-sandbox`: that removes the protection that keeps web pages away from your files.

Advanced: an AppArmor profile for the AppImage

If you must use the AppImage on Ubuntu 23.10 or later, an administrator can allow it. Create the file `/etc/apparmor.d/ferna-sundaree-appimage` with this content (adjust the path to where you keep the file):

```
abi <abi/4.0>,
include <tunables/global>

profile ferna-sundaree-appimage /home/*/Applications/FernaSundaree-*.AppImage flags=
(unconfined) {
  userns,
  include if exists <local/ferna-sundaree-appimage>
}
```

Then load it with `sudo apparmor_parser -r /etc/apparmor.d/ferna-sundaree-appimage` and start the AppImage again.

CHAPTER 5

Use the .tar.gz archive

1. Unpack the archive into a folder of your choice:

```
mkdir -p ~/.local/opt
tar -xzf ~/Downloads/FernaSundaree-0.6.2-x64.tar.gz -C ~/.local/opt
cd ~/.local/opt/FernaSundaree-0.6.2-x64
./ferna-sundaree
```

2. On Ubuntu 23.10 and later, set up Chromium's sandbox helper once, so the sandbox works without an AppArmor profile:

```
sudo chown root:root chrome-sandbox
sudo chmod 4755 chrome-sandbox
```

   Run these inside the unpacked folder. Other distributions do not need this step.

3. To add a menu entry and make it your default browser, open *Menu* → *Settings* → *General* and click **Make default**.

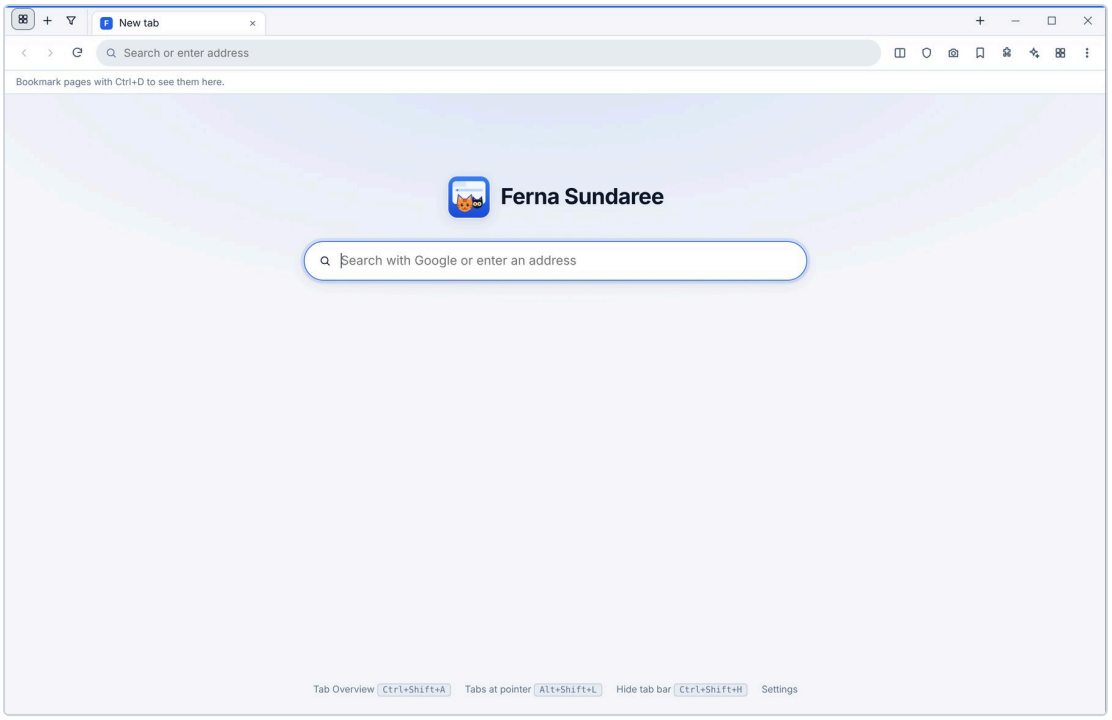
On ARM computers the folder is `FernaSundaree-0.6.2-arm64`. It contains the `ferna-sundaree` program and the `chrome-sandbox` helper.

CHAPTER 6

After installing

The first start

Ferna Sundaree opens with a new tab. On the first start it downloads the ad-blocking and safe-browsing lists in the background. Blocking becomes fully active within a minute.



Ferna Sundaree on its first start. Type an address or a search, or press Ctrl + L to use the address bar.

Make it your default browser

Open **Menu (⋮) → Settings → General** and click **Make default**. Ferna Sundaree becomes the default for web links, HTML files and PDF files for your account. You can also do this in a terminal:

```
xdg-settings set default-web-browser ferna-sundaree.desktop
```

Some desktops have their own setting as well, for example GNOME *Settings → Apps → Default apps* or KDE *System Settings → Default Applications*.

Bring your bookmarks and passwords

- **Bookmarks:** *Menu → Bookmark manager* (Ctrl + Shift + O) → **Import…**. Accepts HTML bookmark files from any browser, Chrome/Chromium/Brave `Bookmarks` files (in `~/.config/google-chrome/Default/` and similar folders) and Firefox `.json` backups.
- **Passwords:** export a CSV file from your old browser or password manager, then *Menu → Passwords and autofill* → **Import CSV…**. Delete the CSV file afterwards: it contains your passwords as plain text.

CHAPTER 7

System components some features use

Ferna Sundaree uses standard Linux components for a few features. Full desktops (Ubuntu, Fedora Workstation, Mint, KDE) usually have the first two already.

Feature	Uses	Install if missing
Encryption of passwords, history, bookmarks	Your login keyring through the Secret Service (GNOME Keyring or KWallet)	`sudo apt install gnome-keyring` (or KWallet on KDE). Without a keyring, the Passwords page says the vault is not encrypted.
Password check before showing or exporting a password, and package updates	polkit (`pkexec`) with your desktop's password dialog	Ubuntu 24.04: `sudo apt install pkexec` ; Debian 12: `policykit-1` ; Fedora: `polkit` . Minimal window managers also need a polkit agent, for example `lxpolkit` or `polkit-gnome` .
Virus scan of downloads	ClamAV (`clamdscan` , or `clamscan`)	Ubuntu/Debian: `sudo apt install clamav clamav-daemon` Fedora: `sudo dnf install clamav clamd clamav-update` , then `sudo freshclam`
Read aloud	speech-dispatcher voices	`sudo apt install speech-dispatcher espeak-ng`
Security keys (passkeys)	USB/NFC FIDO keys or a phone	Works with current systemd. On older systems: `sudo apt install libu2f-udev` , then plug the key in again.
Private window with Tor	Tor Browser (port 9150) or the Tor service (port 9050)	`sudo apt install tor` , or keep Tor Browser open

> **ClamAV needs current virus definitions**
> After installing ClamAV, the `clamav-freshclam` service downloads the definitions and keeps them current. The first download can take a few minutes. Scanning is fastest when `clamav-daemon` is running; otherwise each scan starts the slower `clamscan` .

CHAPTER 8

Updates

Ferna Sundaree checks for a new version a minute after it starts and then every six hours. Chromium security fixes are released about once a week, so expect a small update most weeks. Updates come from GitHub and the Ferna update mirror, whichever has the newest correctly signed release.

Every release carries a list of its files with their checksums, signed with the Ferna project's private key. The browser contains the matching public key. An update is installed only if the signature is valid, the version is newer, and the downloaded file matches the signed checksum.

Package	What happens
.deb	Downloaded in the background. When it is ready, a message says so and the menu shows **Restart to update**. Your desktop asks for your password (polkit), `dpkg` installs the package and the browser restarts.
.rpm	The same, with `rpm -U`.
AppImage	The checked new AppImage replaces your file in place, and the browser restarts. The folder holding the AppImage must be writable by you.
.tar.gz	*Settings* → *About* shows that an update is available. Click **Download update**, then **Save new version**: the archive is saved in Downloads. Unpack it as in Chapter 5.

Open **Menu** → **Settings** → **About** and click **Check now** to look for a new version immediately. Switch **Update automatically** off if you prefer to update by hand. To update by hand, install the newer package in the same way as the first time. Your data stays in place. If you cancel the password dialog, the update waits; choose **Restart to update** again later.

CHAPTER 9

Uninstall

Package	Command
.deb	`sudo apt remove ferna-sundaree`
.rpm (Fedora)	`sudo dnf remove ferna-sundaree`
.rpm (openSUSE)	`sudo zypper remove ferna-sundaree`
AppImage	Delete the AppImage file and, if you made it the default, `~/.local/share/applications/ferna-sundaree-local.desktop`
.tar.gz	Delete the unpacked folder (and the launcher above, if you created one)

Remove your data as well

Uninstalling keeps your bookmarks, passwords, history and settings. To remove them for good, delete these after uninstalling:

```
rm -rf ~/.config/FernaSundaree                          # profiles, passwords, settings
rm -f ~/.local/share/applications/ferna-*.desktop       # launchers for web apps and
profiles
rm -f ~/.local/share/icons/hicolor/256x256/apps/ferna-app-*.png
```

Launchers for web apps and profiles may also have been copied to your desktop. Screenshots you saved stay in `~/Downloads/Ferna Screenshots`.

> **This cannot be undone**
>
> Deleting `~/.config/FernaSundaree` removes saved passwords permanently. Export them first (*Passwords and autofill → Export CSV…*) if you might need them.

CHAPTER 10

For schools and organisations

Managed settings (policies)

Administrators can enforce settings for every user of a computer with JSON files. The setting names are the same as the Windows Group Policy template.

1. Create the folder and copy the example:

```
sudo mkdir -p /etc/ferna-sundaree/policies/managed
sudo cp "/opt/Ferna Sundaree/resources/policy/policy.example.json" \
        /etc/ferna-sundaree/policies/managed/school.json
```

2. Edit the file with `sudo nano /etc/ferna-sundaree/policies/managed/school.json`.
3. Make sure only root can change it, otherwise Ferna Sundaree ignores it:

```
sudo chown root:root /etc/ferna-sundaree/policies/managed/*.json
sudo chmod 644 /etc/ferna-sundaree/policies/managed/*.json
```

4. Restart Ferna Sundaree. Settings set by policy show a lock and cannot be changed by users.

Example file:

```
{
  "HttpsFirst": true,
  "AdBlocker": true,
  "SafeBrowsing": true,
  "DefenderScan": true,
  "HistoryDays": 90,
  "TorWindowsAllowed": false,
  "DeveloperToolsAllowed": false,
  "ExtensionInstallAllowed": false
}
```

Several files are read in name order, and a later file overrides an earlier one. Available names: **HttpsFirst, AdBlocker, SafeBrowsing, SearchEngine, DnsOverHttpsMode, DnsOverHttpsProvider, BlockThirdPartyCookies, EnhancedSecurity, AutoUpdate, PasswordManager, AutofillAddresses, AutofillCards, LookalikeWarnings, DefenderScan** (ClamAV on Linux), **ExtensionGuard,**

FingerprintProtection, GlobalPrivacyControl, PopupBlocker, SleepingTabs, HistoryDays, UpdateMirror (your own update folder), and the feature switches **AssistantAllowed, SyncAllowed, PrivateWindowsAllowed, TorWindowsAllowed, DeveloperToolsAllowed, WebAppsAllowed, ProfilesAllowed** (true = allowed). *ExtensionInstallAllowed* is accepted but not enforced yet.

Deploying to many computers

Copy the .deb or .rpm and the policy file with your usual tools (Ansible, Puppet, Salt or a shell script) and install with `apt install ./…deb` or `dnf install ./…rpm`. Set `"AutoUpdate": false` if you roll out new versions yourself.

Exam computers

Secure exam mode works on Linux: a full-screen, always-on-top window with only the allowed sites, ended by the invigilator's password or the time limit. **One difference from Windows:** Linux cannot hide a window from screenshot and screen-sharing programs. Remove or disable such programs on exam computers, and use standard (non-sudo) student accounts.

CHAPTER 11

Troubleshooting

Problem	What to do
The AppImage or .tar.gz stops with a message about the **sandbox** (Ubuntu 23.10+)	Use the .deb (Chapter 2). For the .tar.gz, set up `chrome-sandbox` (Chapter 5). Do not use `--no-sandbox`.
The AppImage does nothing / FUSE error	Install FUSE 2 (Chapter 4), or run it with `--appimage-extract-and-run`.
“The vault is stored without encryption”	No keyring is running. Install and unlock GNOME Keyring or KWallet, sign out and back in.
No password dialog appears when showing a password or updating	Install polkit and a polkit agent (Chapter 7).
Downloads show no “ClamAV: no threats” line	ClamAV is not installed or has no virus definitions yet (Chapter 7).
Blank or flickering window (often Wayland with NVIDIA drivers)	Start with `ferna-sundaree --ozone-platform=x11`. If that helps, add the option to the menu entry, or sign in to an X11 session.
apt: “unable to locate package”	Put `./` before the file name, and run the command in the folder that holds the file.
Wrong architecture error	You downloaded x64 for an ARM computer or the other way round (Chapter 1).

Start completely fresh	Close the browser and rename `~/.config/FernaSundaree/Profiles/Default` to `Default.old`.

Command-line options

Option	Effect
`--profile=Work`	Starts the browser with the profile *Work*, which has its own bookmarks, passwords, cookies and extensions
`--app=https://example.com`	Opens a site in its own app window
`--exam=/path/quiz.fernaexam`	Starts a secure exam from an exam file
`--ozone-platform=x11`	Uses X11 instead of Wayland
an address or a file path	Opens it in a new tab

Help and bug reports: github.com/tgi25/ferna-sundaree/issues. Please mention your distribution and version, your desktop, the package type and the Ferna Sundaree version from *Settings → About*.